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In vitro post-meiotic germ cell development from human embryonic stem cells

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BACKGROUND: Investigating the mechanisms of human primordial germ cell (PGC) and gamete development are important for understanding the causes of infertility and effects of environmental chemicals on reproductive development. However, there are practical and ethical difficulties associated with obtaining human tissue in early development. The aim of this study was to investigate whether human embryonic stem cell-hESC-generated germ cells could provide an *in vitro* model of gamete development.

METHOD: Human ESCs were differentiated as embryoid bodies (EBs) *in vitro*. Gene and protein marker expression profiles of EBs in different periods of culture were analysed by quantitative polymerase chain reaction (Q-PCR) and immunolocalization to monitor germ cell development. Secretion of hormones involved in germ cell maturation was measured, to detect the existence of a germ cell niche within EBs.

RESULTS: Q-PCR revealed gene expression profiles consistent with PGC formation and germ cell development. A small population of post-meiotic spermatid cells were identified using sperm-specific antibodies (Protamine 1 and 1.97). Although gene expression profiles characteristic of oocyte development and follicle-like structures were detected, a committed oocyte with extra-cellular zona pellucida was not recognized with zona pellucida-specific monoclonal antibody.

CONCLUSIONS: hESCs can form PGCs and post-meiotic spermatids *in vitro*, however, there remains doubt about oocyte development. Levels of steroid hormones produced by EBs were significant when compared with known values for a similar quantity of human testis, suggesting that hESC may intrinsically create a favourable hormonal niche for spermatogenesis.

Key words: oocyte / germ cells / human embryonic stem cells / niche / sperm

Introduction

Elucidation of the molecular mechanisms of germ cell specification and gamete development is crucial for understanding epigenetic and reprogramming phenomena of the germ cell cycle and gametogenesis, and the factors that might disrupt these processes. Environmental toxicants such as fungicides may have profound *trans*-generational effects on mammalian germ cell function leading to infertility and cancers (Anway *et al*., 2005; Tiido *et al*., 2005). Furthermore, there is increasing evidence that many testicular cancers have an origin in aberrant development of the primordial germ cell (PGC) and gonocyte (Rajpert-De Meyts, 2006; Rajpert-de Meyts and Hoei-Hansen, 2007). Although, we have gained considerable knowledge of early germ and gamete development in the mouse and many mechanisms are highly conserved in mammals (Donovan and de Miguel, 2003; McLaren, 2003), there remains a need to investigate human germ cell development directly (Moore and Aflatoonian, 2007). Therefore, *in vitro* culture systems of human germ cells are of significant value and their use may lead to novel methods to monitor the effects of reproductive toxicants and eventually novel clinical applications to treat infertility.

First trimester human fetal gonadal tissue can be a source of primary tissue, but there is a paucity of samples for the earliest stages of development. However, embryonic stem cells may offer a sustainable source of cells to investigate germ cell development *in vitro*. There is good evidence that murine embryonic stem cells (mESCs) in permissive culture can differentiate to PGCs and a small population may undergo further development and meiosis to immature gametes, which after fertilization apparently form blastocysts

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(Hubner *et al*., 2003; Toyooka *et al*., 2003; Geijsen *et al*., 2004). Moreover, it has been reported that immature sperm cells derived from mESCs in culture have generated live offspring (Nayernia *et al*., 2006). Human embryonic stem cells (hESCs) also might display a similar developmental capacity (Clark *et al*., 2004; Aflatoonian *et al*., 2005). Under a variety of conditions involving spontaneous (Clark *et al*., 2004; Aflatoonian *et al*., 2005) or growth factor-induced culture (Kee *et al*. 2006; Tilgner *et al*., 2008; West *et al*., 2008; Park *et al*., 2009), hESCs differentiate and express protein markers and gene expressions profile consistent with germ cells. Although PGC development can be inferred from these profiles, post-meiotic cell phenotypes have been difficult to identify and recover. The most important event determining the nuclear status of spermatozoa is the replacement of histones that surround the chromatin just after meiosis by protamines (PRM1 and PRM2). Transition nuclear proteins (TNP1 and TNP2) are thought to promote histone–protamine exchange (Steger *et al*., 1998; Lee *et al*., 2001). Another specific marker during spermatogenesis is testis-specific histone 1 (H1t). H1t mRNA has only been detected in the cytoplasm of mid and late pachytene spermatocytes (Steger *et al*., 1998). Moreover, previously in our laboratory, a monoclonal antibody (1.97) was generated as a perinuclear specific marker for the human sperm head (Al-Eisa *et al*., 2001) which was shown to be highly specific to human sperm and did not recognize other tissues. For oogenesis, zona pellucida proteins are perhaps the best markers of post-meiotic development, although growth and differentiation factor9 (GDF9, BMP15) has often been used as a marker.

Here, we investigate the *in vitro* differentiation of human PGCs and generation of an embryonic germ (EG) cell line from these cells; demonstrate that a hormone environment conducive for spermatogenesis can be established following embryoid body (EB) formation; and identify post-meiotic early spermatid development. In contrast, oocyte development could not be confirmed by immunolocalization with specific monoclonal antibody to zona pellucida.

Materials and Methods

HESC lines and maintenance

Shef 1, 3, 4, 5 and 6 cell lines were generated by the Centre for Stem Cell Biology (CSCB) under license from the Human Fertilization and Embryology Authority. These hESC lines have been verified and deposited in the UK Stem Cell Bank for international distribution. All Shef cell lines display marker profiles of pluripotency (Innis and Moore, 2006; Adewemi *et al*., 2007). The H7 hESC cell line was obtained as a gift from Dr J. Thomson, University of Wisconsin (initially to Professor Peter Andrews, CSCB). All six hESC lines were chromosomally normal at the beginning and end of the study, as tested by karyotype analysis using the G-banding technique. Cells were maintained as previously described under standard protocols using inactivated mouse embryonic fibroblasts (MEFs) with supplemented Knock-Out DMEM medium (Invitrogen) at 37°C at 5% CO_2 in air (Draper *et al*., 2004).

HESC differentiation cultures

To induce spontaneous differentiation, HESCs were dissociated by incubation with collagenase type IV (1 mg/ml) and cultured in EB medium consisting of hESC medium without basic fibroblast growth factor (bFGF). Non-adherent cell cultures were prepared by transferring 12 ml of cell suspension into bacteriological (non-tissue culture treated) Petri dishes and incubating at 37°C in 5% CO_2 in air (Draper *et al*., 2004). Half of the medium was changed every 2 days. Several induction conditions for germ cell development from EBs were also used. (i) EB medium was supplemented with bone morphogenic protein 4 (BMP4 10 ng/ ml; R&D systems); (ii) EB medium was supplemented with Retinoic acid (RA; 2 μM; Sigma); (iii) Conditioned medium from neonatal mouse testis (Lacham-Kaplan *et al*., 2006). Neonatal mouse testes (NMT) from MF1 strain of mice were cultured in DMEM medium supplemented with 10% fetal calf serum (FCS) and 20 μg/ml gentamycin. Every 5 days the medium was collected as conditioned medium and stored at −20°C. This conditioned medium (35 ml) was added to EB medium to make 100 ml, filtered and used as NMT conditioned medium. C-KIT$^+$, SSEA1$^+$ cells isolated by FACS were cultured in EG cell medium consisting of 65 ml of standard HES medium supplemented with 10 ng/ml BMP4, (R&D system), 1000 U/ml of human recombinant leukaemia inhibitory factor (Chemicon), 10 μM forskolin (Sigma), 2 μM RA (Sigma) and 35 ml of NMT condition medium, The medium was filtered using a 225 ml 0.2 μm filter, stored at 4°C and used within 2 weeks. It was not feasible to monitor karyotypes of individual EB cultures but in random samples there was no indication of genetic instability during differentiation. Experiments were performed in triplicate.

RNA isolation, cDNA production and quantitative PCR

HESCs at Day 0 and EBs at Days 4, 7, 14 were collected by aspiration into an Eppendorf tube, centrifuged, the supernatant discarded and 1 ml of TRI Reagent© (Sigma) added to each sample. Tubes were vortexed to completely lyse the cells and stored at −20°C until analysis. Total RNA was extracted using TRI Reagent standard protocol supplied by the manufacturer. Total RNA was treated with DNase I (DNA-free Kit; Ambion, Huntingdon, UK) to remove genomic DNA contamination from samples. First-strand cDNA synthesis was performed using oligo dT primers and the Superscript II reverse transcriptase system (Invitrogen, Paisley, UK). For the genes *OCT4*, *VASA*, *DAZL*, *SCP3* and *GDF9* quantitative PCR (Q-PCR) was performed using the Assay-on-Demand technology (Applied Biosystems) according to manufacturer's specifications using FAM-490. For the genes *NANOG*, *SOX2*, *ZP1*, *H1t*, *TNP1*, *TNP2*, *PRM1* and *PRM2*, Q-PCR was performed using prepared cDNA and primers. Experiments included a negative control with no cDNA and PCR products were sequenced to confirm identity. Forward and reverse primer sequences are shown in Table I. SYBR Green Jump Start (Sigma) master mix was added to each well of the PCR plate (10 μl of SYBR Green, 7 μl of water, 2 μl of primers and 1 μl of cDNA), and PCR was performed under the following conditions: 50 cycles at 95°C for 30 s, 59–70°C for 30 s (different temperature for different primers) and 72°C for 30 s. Samples were run in triplicate. All results were analyzed using an iCycler (Biorad laboratories, Hemel Hempstead, UK). The threshold of cycle values was normalized against the threshold value of Human *β-ACTIN*. Differences in normalized expression values between samples were tested for significance using ANOVA statistical test. The results were expressed as mean ± SEM. The level of statistical significance was set at $P < 0.05$ (Aflatoonian *et al*., 2007).

Immunofluorescent localization

Cells and EBs from culture were washed with PBS containing 1% FCS before fixing in 4% paraformaldehyde in PBS (pH 7.4) for 15 min. Alternatively, EBs were collected and cultured in chamber slides (4-wells) with

Table I List of primers used for Q-PCR showing specifications

Gene	Forward primer (5′–3′)	Reverse primer (5′–3′)	Annealing temp. (°C)	Product size (bp)	Accession number
NANOG	CTGGATTTGTGGGCCTGAA	TGTTTGCCTTTGGGACTGGT	65	153	NM_024865
SOX2	AGAACCCCAAGATGCACAAC	GCTTAGCCTCGTCGATGAAC	65	100	NM_003106
ZP1	TGAGAACTGGCTGGTGTCTG	GCTGAAGGTCTCGTCTTTGG	53	217	NM_207341
H1t	CTTCTGCCAGTGCTGGTGTA	ACACAGAGAGGTTCGGCACT	59	111	NM_005323
TNP1	AAGAGCCGATCTCCTCACAA	ACAAGTGGGAGCGGTAATTG	59	121	NM_003284
TNP2	CAGCCAGAGTTGCAGACAGA	TTCAGCTTGCCTTCCAAGTT	65	243	NM_005425
PRM1	AAGAAGTCGCAGACGAAGGA	TTCTCAGGCAGGAGTTTGGT	53	115	NM_002761
PRM2	CCCATGGCCAGTCTCACTAT	ATTGCTATGGCCTCACTTGG	65	244	NM_002762
ß-ACTIN	GCCAGCTCACCATGGATCAT	CAAACATGATCTGGGTCATCTTC	59	384	NM_001101

DMEM+10% FCS overnight to enable attachment. Samples were washed twice for 5 min in PBS containing 1% FCS followed by incubation in 0.1% Triton X in PBS for 5 min and incubated overnight at 4°C with primary antibody against germline lineage-enriched markers (markers highly expressed in germ cells with limited expression in somatic cells), and specific markers (only expressed in germ cell). Antibodies were diluted in PBS [1/50, C-KIT; 1/100, DAZL; 1/100, VASA (all from abcam, Cambridge, UK); 1/5 SSEA1; 1/5, TRA-1-81; 1/5, TRA-1-60; 1/5, TRA-2-49; 1/5, TRA-2-54 (Draper *et al*., 2004; CSCB); 1/200, PRM1; 1/200 GDF9B (Santa Cruz, Heidelberg, Germany); 1/1, 1.97 (Al-Eisa *et al*., 2001; CSCB); 1/500, 1/1000, 1/5000]. A double staining kit (DakoCytomation Envision Doublestain System) was also used for immunohistochemical localization of DAZL and VASA. A monoclonal antibody against pig oocyte zona-pellucida antigen (Z061) which cross-reacted against human zona pellucida was produced in-house (Aflatoonian *et al*., unpublished).

All primary antibodies were incubated overnight at 4°C. Colonies were then washed twice in PBS and incubated with appropriate secondary antibodies for 1 h in 37°C. Preparations were covered with mounting medium (Vectashield; Vector laboratories, USA) or PBS and examined by microscopy using phase contrast and UV excitation optics.

Flow cytometry

EBs were collected at designated periods of culture washed with 1% FCS in PBS, and treated with trypsin/EDTA for 10 min to form single cells. The suspension was inactivated with 10% FCS in DMEM and cells washed twice by centrifugation before labelling and flow cytometry as described previously (Draper *et al*., 2002). Additionally, EBs of Shef 1 were dispersed to single cells labelled with SSEA1 and c-Kit (markers for PGCs) and sorted using fluorescent activated cell sorter (FACS) for isolation of PGCs. Positive cells ($SSEA1^+/C\text{-}KIT^+$) were cultured on feeder with HES medium for 14 days. To analyse DNA content, single cells were analysed for DNA content using a Dako ADP CyAn with O2 optics (Draper *et al*., 2002). The amount of DNA in each cell was quantified using a 488 nm 20 mW semiconductor laser, with red fluorescence measured using a 613/20 nm band pass and a 640 nm long pass dichroic filter. Doublet discrimination was undertaken using linear peak intensity versus peak area and the DNA content was displayed on as linear peak intensity frequency histogram.

Measurement of hormones (testosterone, dihydrotestosterone, estradiol) from EB conditioned media

Four batches of EBs (Shef 4) were cultured in four different conditions; EB medium; EB+10 ng/ml BMP4; EB+2 μM RA; EB+NMT. Medium was changed (50%) every 2 days. One millilitre of medium was collected after 4, 7, 14 days, and stored at −20°C. Two batches of this collected conditioned media plus 1 ml of each different stock medium (as control) were analysed by sensitive enzyme-linked immunosorbent assay (ELISA; courtesy of Dr Jenny Jones, King's College London, School of Medicine, University of London).

Results

EB generation and differentiation

For spontaneous EB development, the Q-PCR mRNA profiles for genes involved at various stages of PGC/gametogenesis are shown in Fig. 1. *OCT4* and *NANOG* mRNAs were highly expressed in hESCs at Day 0 and decreased after 4, and 7 days but were still detectable at Day 14, albeit at low levels. *DAZL* and *VASA* displayed significantly elevated expression at Day 4 and 7, respectively. The peri-meiotic marker *SCP3* displayed a progressive increase to Day 14. For spermiogenesis there were significant increases in post meiotic mRNA markers over culture with transitions protein expression elevated as early as 4 days for TNP1. Although *PRM2* was detected early there was no appreciable detection of *PRM1* before 14 days. In contrast, expression of the oogenesis markers *GDF9* and *ZP1* increased progressively over the period of culture.

Under the various induced conditions for gametogenesis, only medium with retinoic acid gave significantly elevated expression of *VASA* at Day 7 of EB culture and *PRM1* at Day 14 (Fig. 2).

After Day 4, most (>80%) EBs did not display the hESC pluripotency marker SSEA3 although they contained a variable proportion of cells displaying the markers TRA-1-60, SSEA1 and liver specific alkaline phosphatase (TRA-254) as also expressed by PGCs. The proportion of cells in an EB displaying germ cell markers varied markedly from 10 to 80%. Other germ cell markers expressed in

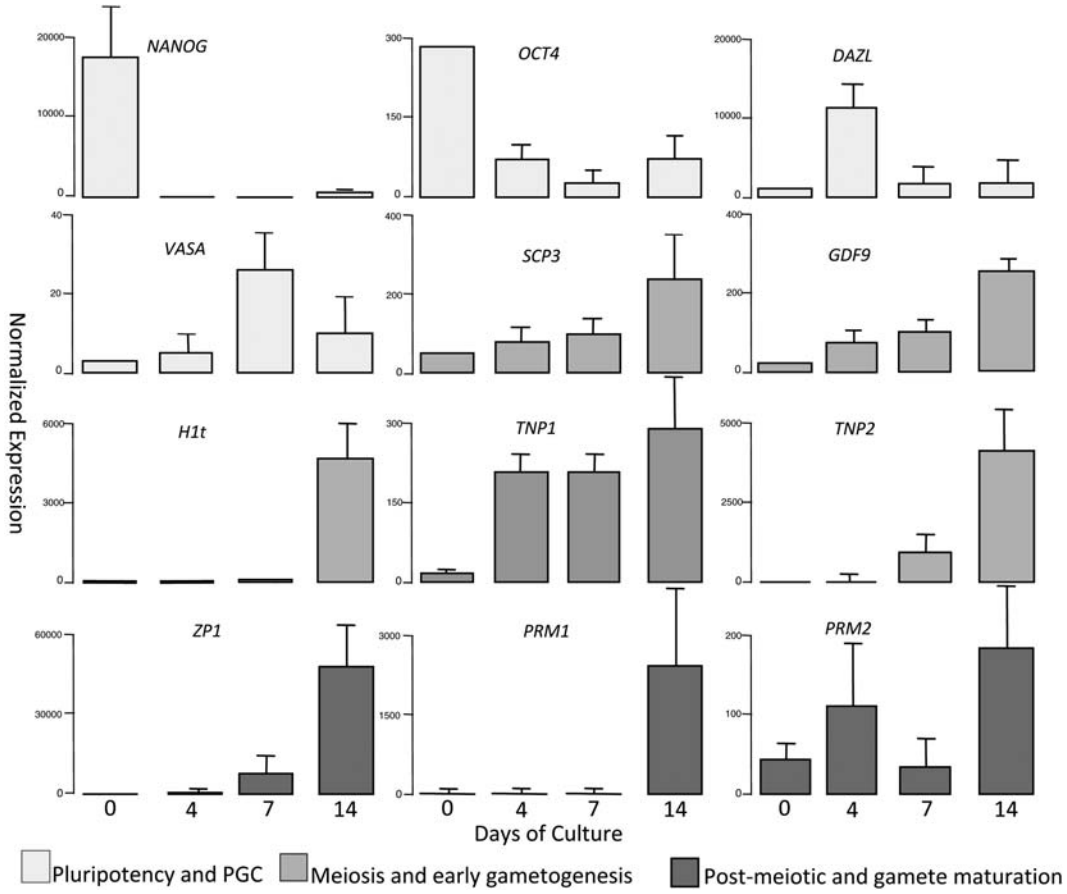


Figure 1 Q-PCR analysis of normalized expression of key genes involved in embryoid body spontaneous differentiation of germ cell development over 14 days of culture.

Error bars are standard errors of the mean. PGC, primordial germ cell.

EBs were C-KIT (Fig. 3A), VASA, (Figs. 3C and 4A and B) and DAZL (Figs. 3E and 4C). Significantly, immunofluorescent or histochemical localization identified ribbons or cord-like structures of cells at Days 7 and 14 (Figs. 3E and 4A and B).

EBs of Shef 1 were dispersed to single cells, labelled with SSEA1 and C-KIT (markers for PGCs) and sorted by FACS for isolation of putative PGCs. A small proportion (0.5–1%) of cells positive for both markers (Fig. 5) were selected, isolated and cultured on feeders with HES medium. After 3 weeks, two colonies were identified, isolated and underwent further proliferation in modified EG medium as putative EG cell lines.

At Day 14 (but not before), a small proportion of cells (1–5%) in EBs expressed specific protein markers of spermiogenesis. Single cells were labelled intensely with monoclonal antibodies to either Protamine 1 (PRM1; Fig. 3G and H) or the post-meiotic peri-nuclear sperm marker 1.97 (Al-Eisa *et al.*, 2002; Fig. 6). Moreover co-localization with C-KIT and 1.97 antibodies were performed (Fig. 7). These cells had morphology very similar to a round spermatid with a compacted nucleus (Fig. 7) and most showed indications of the beginning of flagellum formation with a fine axial filament reminiscent of round mouse spermatid development in culture for 20 h (Gerton and Millette, 1984). When EBs were subsequently dispersed as single cells in monolayer and then analysed for DNA content, a small population of cells with a 1 N (haploid) chromosome DNA content was detected at Day 14 of culture (Fig. 8).

In contrast, although cellular morphology closely resembling oocytes and primordial follicles was observed in EBs, it was not possible to obtain immunolocalization of zona pellucida matrix (Fig. 9).

The steroids, testosterone (T), dihydrotestosterone (DHT) and estradiol (E_2) were not detected in control medium except for NMT conditioned medium where DHT was 5–20 pg/ml. Levels of DHT increased in all conditioned medium to 52 ± 15 pg/ml in EB culture for 14 days (50% replacement of medium every 2 days; 5 ml medium per flask) with highest levels in NMT conditioned (60–70 pg/ml). E_2 was only detected in NMT conditioned medium (22 ± 8 pg/ml).

Discussion

Our results confirm previous findings shown for mouse ESCs (Geijsen *et al.*, 2004; Toyooka *et al.*, 2003; Nayernia *et al.*, 2006) and hESCs (Kee *et al.*, 2006; Tilgner *et al.*, 2008; West *et al.*, 2008), and indicate

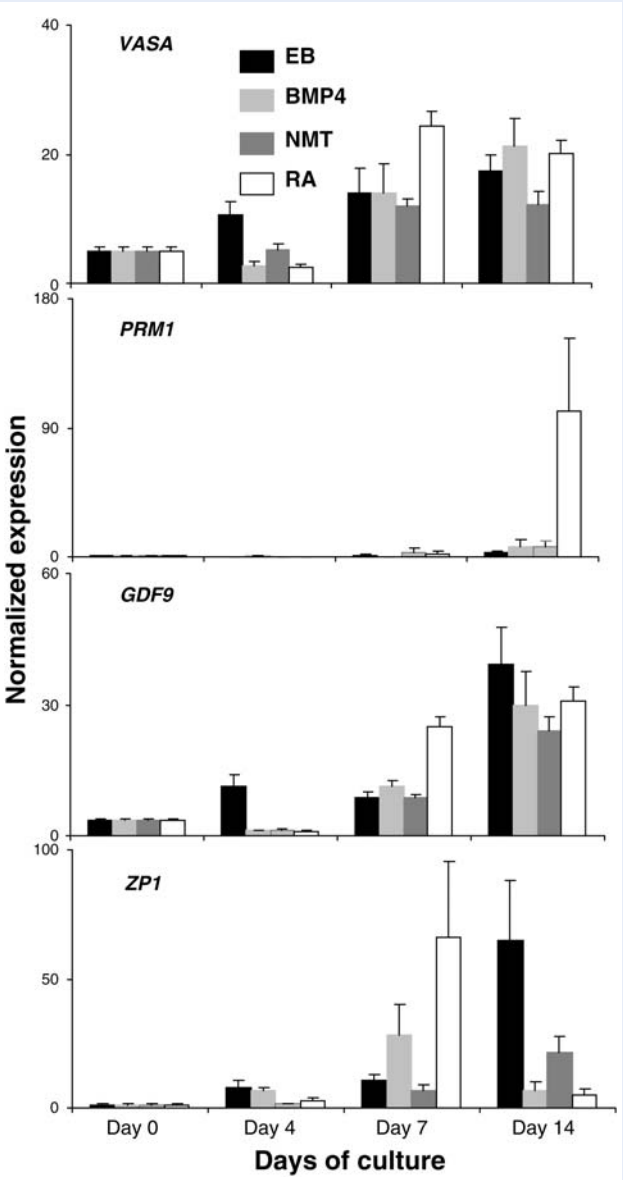


Figure 2 Q-PCR analysis of normalized expression of selected genes involved in embryoid body induced differentiation of germ cell development with various factors and supplements.

Error bars are standard error of the mean. EB, embroid body media; BMP-4, bone morphogenetic protein 4 conditioned media; NMT, neonatal mouse testes conditioned media; RA, retinoic acid conditioned media.

that hESCs differentiating in suitable culture conditions may undergo germ cell development although the extent of gametogenesis is limited. In the present study, we observed cells with a reduced haploid DNA complement albeit in this case representing a very small proportion of the total cell population in EBs (~0.5%). But more significantly, spermatid-like cells with condensed chromatin (with some displaying a short immature flagellum or axial filament) and sperm-specific protein expression were consistently identified suggesting an occasional post-meiotic phenotype. By 14 days, the expression of SCP3, as a peri-meiosis marker, and dual immunostaining of a small number of spermatid-like cells as PRM1$^+$ or C-KIT$^+$/1.97$^+$ (recognising a specific perinuclear theca antigen) most probably indicated the beginning of post-meiotic spermiogenesis. Most of these cells displayed the beginning of a fine axial filament representing the first stage of flagellum development. This development has been shown previously in culture with mouse spermatids with the filament extending over a 20 h period (Gerton and Millette, 1984). Formation of spermatid-like cells was not obviously related to the karyotype of the hESCs as they were generated in EBs from genetic male (Shef 1, 3, 4) and female (Shef 5, 6) cell lines in keeping with observations that human germ cells differentiated *in vitro* may expressed both the male and female genetic programmes regardless of sex karyotype (Clark *et al*., 2004; Lacham-Kaplan, *et al*., 2006). Previously it was reported that by Day 14 of human EB culture, SCP3 protein was detected in some cells predominantly in the cytoplasm but also in punctate SCP3-positive nuclear structures (Clark *et al*., 2004) although another meiosis marker MLH1 was not detected, nor was the formation of haploid germ cells observed. By contrast, Tilgner and coworkers (2008) detected an apparently small haploid population of cells as determined by DNA content using flow cytometry but were unable to identify cells with functional synaptonemal complexes by immuno-cytochemical analysis, although SCP3 gene expression was detected. The above studies detected only a relatively small proportion of cells displaying peri-meiosis markers. But it has also been reported (West *et al*., 2008) that when hESCs are cultured for 16 days in a continuous monolayer with mouse feeders in medium supplemented with bFGF, 90% become positive for MLH1 and SYCP3 protein that staining was localized to the nucleus. This suggests that cells might be preparing for meiosis. These reported differences in germ cell development may in part reflect differences in the gametogenesis capacity of the different cell lines used under the various permissive conditions.

After 14 day EB culture, it proved impossible to isolate spermatid-like cells as a viable population from collagenase and trypsin digests due to the amount of extracellular matrix material. However, viable surface-labelled SSEA1$^+$/C-KIT$^+$ single cells were recovered by FACS after 4 days. These cells formed a monolayer in culture and if subsequently analysed for DNA complement displayed a small but consistently discernible haploid population after 14 days total culture. Although these cells displayed a condensed nucleus they did not show the more obvious spermatid morphology of the cells in 3-D development in EBs. In order to try and capture a PGC progenitor population the SSEA1$^+$/CKIT$^+$ sorted cells were cultured further with MEF feeders in hES medium and two colonies of cells developed. These were manually passaged and transferred to an EG induction medium where hEG-like cells proliferated. The medium was modified from the standard medium used to initially derive hEG cells (Shamblott *et al*., 1998) by adding BMP4-, RA- or NMT-conditioned medium.

Germ cells will enter meiosis autonomously and develop as oocytes unless meiosis is blocked and cells are induced into a spermatogenic pathway. Therefore one might expect oogenesis or the generation of oocyte-like cells to be the more likely differentiation route than spermiogenesis. On the basis of light microscopy (bright and phase-contrast), structures were present consistently in EBs at 14 days that closely resembled a primordial follicle. Others have noted this observation for murine and hESC differentiation (Hubner *et al*., 2003; Lacham-Kaplan *et al*., 2006). The differentiation of these putative follicles also correlated with expression of mRNA for the

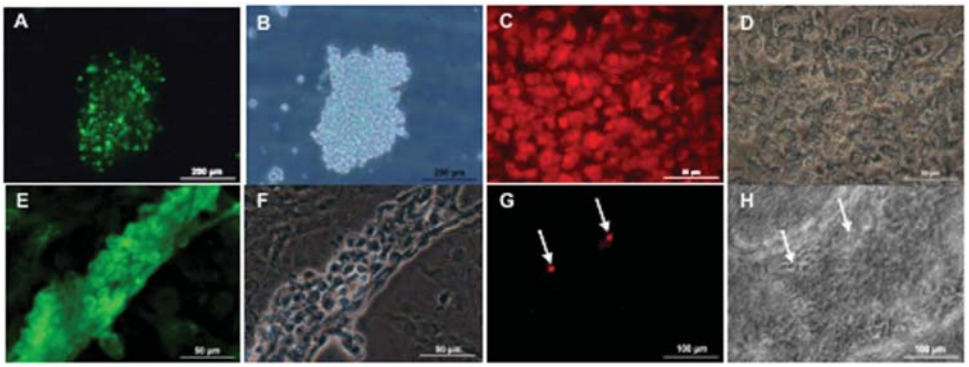


Figure 3 Immunofluorescent localization of markers for PGCs and germ cells in derived from embryoid bodies.
(**A**) C-KIT (Day 4), (**C**) DAZL (Day 4), (**E**) VASA (Day 7) and (**G**) PRM1 (Day 14). Corresponding phase-contrast micrographs, **B, D, F** and **H,** respectively.

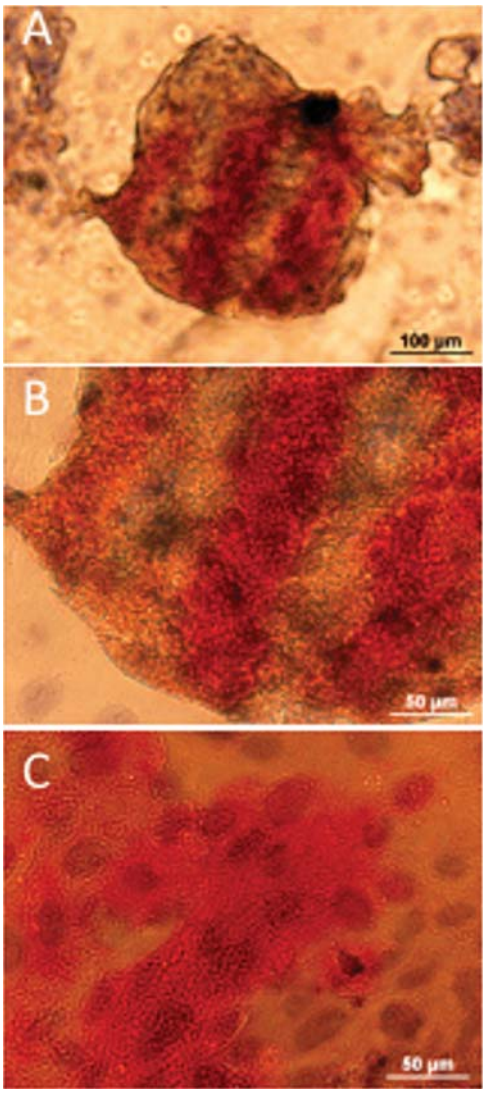


Figure 4 Immunohistochemical localization (red stain) of VASA (**A** and **B**) and DAZL (**C**) in sex cord-like structures of embryoid bodies after 7 days in EB culture medium.

oogenesis markers GDF9 and ZP1. However, in these cultures, there was a failure to detect zona pellucida antigen even though monoclonal antibody (total of six different cloned hybridomas) clearly detected human zona pellucida at all stages of development *in situ*. One explanation is that the zona pellucida antigenic moieties detected in human oocytes by the monoclonal antibodies were absent or aberrant in these follicle-like structures and therefore not detected. Alternatively, identification of a cell /tissue type based on a correlation of total gene expression pattern and morphological features alone is very unreliable for hESC differentiation *in vitro* where many different cellular configurations can develop and the structures observed were not oocytes or follicles. Interestingly, the promoter region of the murine *Gdf9* coupled to enhanced green fluorescent protein (eGFP) has been stably transfected into XX mouse ESCs (Salvador *et al*., 2008). Although eGFP is expressed only in oocytes of chimeric mice generated from the transfected XX ESCs when transfected ESCs were cultured on feeder layers large eGFP-positive cells, surrounded by a structure resembling a zona pellucida appeared transiently on Days 1 and 2. However, in EB cultures, eGFP-positive cells appeared transiently and then re-appeared in regional clusters after a relatively long culture of 30–45 days.

We initially adopted an EB differentiation process because it was considered that 3-D culture would more closely mimic conditions *in situ* than monolayer culture. This was borne out by the immunolocalization of sex cord-like structures at 7 days of culture (Fig. 2) and in PGCs detected as ribbons of cells close to the periphery of EBs. The level of mRNA expression for various marker genes of PGC and germ cells generally followed a profile consistent with development of germ cells although compressed within a relatively short time period when compared with *in vivo*, as has also been observed by several groups (Clark *et al*., 2004; Tilgner *et al*., 2008; West *et al*., 2008). Expression of VASA, which is a marker for gonocytes was at the highest level after 7 days whereas mRNA of genes specifically involved in the transition to spermiogenesis such as H1t, PRM1 and 2, TNP1 and 2 were elevated at 14 days.

Clearly, initial PGC development can occur spontaneously for hESCs in a monolayer or as EBs, as established in the present study. Although several different culture conditions were investigated for inducing germ cells for one cell line (Shef 4) the expression of key genes displayed a variable response. However,

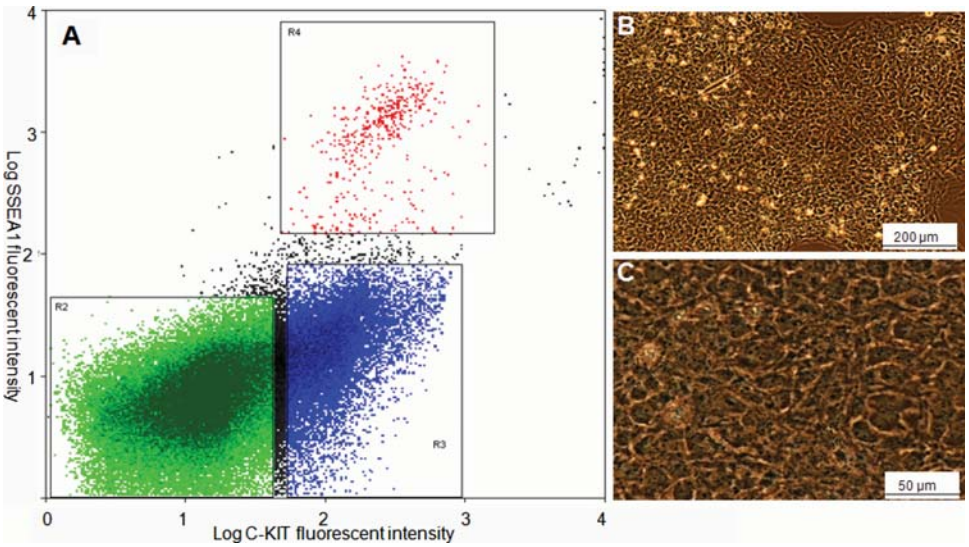


Figure 5 Isolation of putative PGCs by flow cytometry and electrostatic sorting.

(**A**) Embryoid bodies from hESCs (SHEF1) were labelled with antibodies to SSEA1 (*Y* axis) and C-KIT (*X* axis) for flow cytometric analysis and sorting in sterile condition. R2 shows population of cells which were negative for both C-KIT and SSEA1. R3, identifies cells positive for C-KIT; R4, cells positive for both C-KIT and SSEA1. A small proportion (0.5–1%) of cells positive for both markers were selected, isolated and cultured on feeders with HES medium. After 3 weeks, two colonies (**B** and **C**) were identified, isolated and underwent further proliferation in modified EG medium as putative EG cell lines.

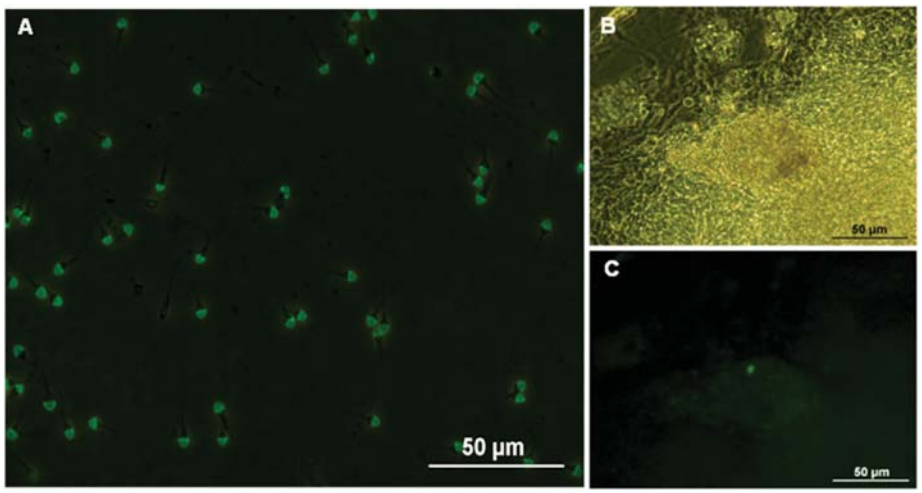


Figure 6 Expression of a sperm-specific marker within embryoid bodies.

(**A**) Localization of 1.97antigen on human sperm head, positive control. (**B**) Phase contrast of EB and corresponding localization of 1.97 in small group of cells with occasional bright fluorescence (condensed sperm heads).

supplementation of EB medium with RA substantially increased expression of PRM 1 at Day 14 compared with other medium. RA is known to bring about rapid differentiation of hESCs in culture to various cell lineages but also plays a key role in early gametogenesis (Bowles *et al.*, 2006; Koubova *et al.*, 2006) and it has been proposed that instead of an intrinsic programme to enter meiosis, germ cells respond to the external signal of RA and its metabolism. Thus, the induction of presumptive PGCs into meiosis in culture medium containing RA might be expected although local concentrations within cell aggregates could differ significantly and affect the timing. Concentrations of RA in plasma or serum (or serum replacement) that supplement hESC culture medium are normally in the 2–8 nM range (Eckhoff and Nau, 1990). We supplemented medium with 2 µM RA (a level often

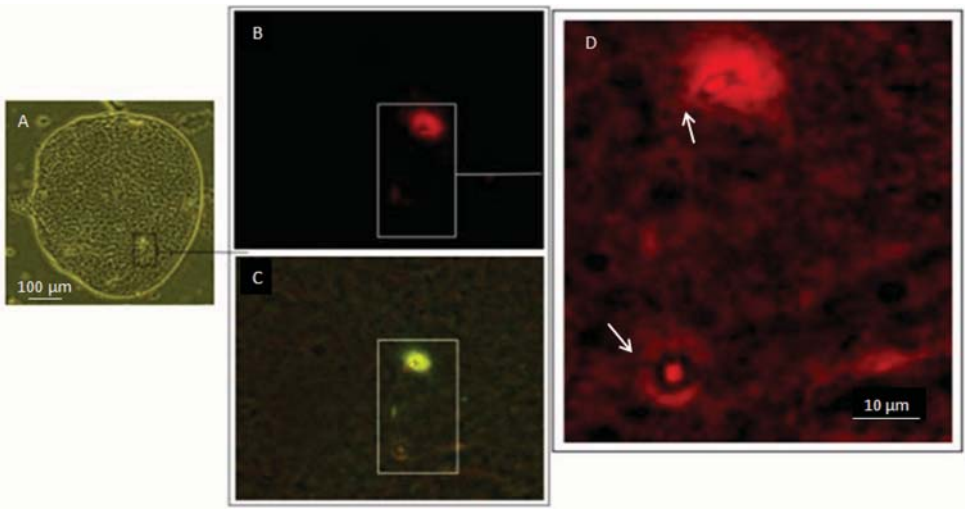


Figure 7 (**A**) Mounted and squashed embryoid body following dual immunolocalization for C-KIT (**B**) and 1.97 antigen (**C**). At higher magnification (**D**) the cells had a condensed early spermatid morphology with indications of flagella development (arrowed).

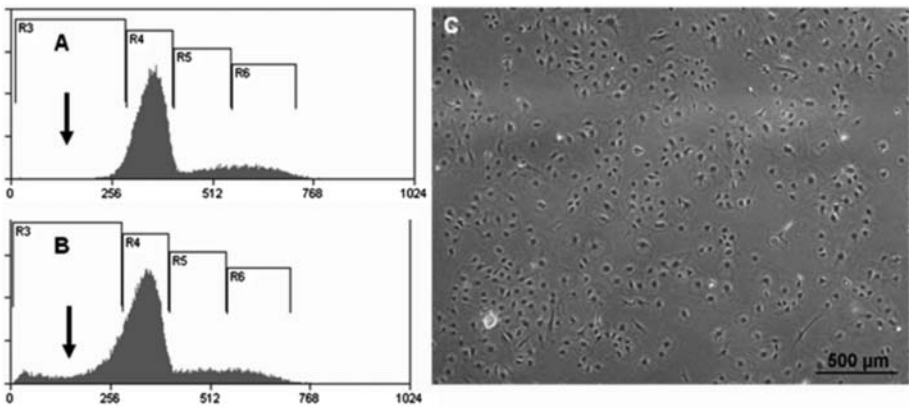


Figure 8 EBs after 7 and 14 days dispersed as single cells in monolayer and analysed for DNA content.

Whereas no 1 N cells were detected at 7 days (**A**) a small population of 1 N cells were detected at day 14 of culture (**B**). A small number of spermatid-like cells with highly condensed nuclei (bright under phase contrast) were observed (**C**).

used with induction protocols) since the sensitivity of cells to RA in culture can vary considerably depending on conditions.

Recently Silva and co-workers reported that a combination of RA and testosterone was conducive for induction of mouse ESC to germ cell development (Silva *et al*., 2009). The conditioned media of our EB cultures contained detectable levels of DHT and E_2. Testosterone (T) was not detectable above background but this hormone can be rapidly converted irreversibly to DHT in the presence of 5α-reductase activity or can be aromatized to E_2. Both DHT and estrogens are crucial for spermatogenesis (Sharpe, 1993). Levels of DHT increased in EB culture over 14 days even though 50% medium was replaced every 2 days. Furthermore, centrifuged pellets of EBs from culture flasks were usually only ∼50 μl in volume therefore if the w/v is 1:1, total DHT secretion was estimated to be 5 (ml of medium) × 50 (pg/ml concentration)/ 50 μg EB tissue = 5 ng/mg tissue. Human testicular levels of DHT have been estimated to be about 13 ng/ml of testicular fluid (Jarrow and Zirkin, 2005), and therefore it quite conceivable that some EB cultures might reach a local level of androgen concentration consistent with spermatogenesis. Whether such steroid originates from a differentiation testicular cell phenotype (i.e. Leydig cell) or from incidental steroid production from non-testicular cells (adrenal, liver phenotypes) was not determined. Certainly EBs readily secretes hCG via spontaneous syncytiotrophoblast differentiation (Harun *et al*., 2006; Moore *et al*., 2008) and therefore will prime any Leydig LH receptors for steroid production. We are currently undertaking further mass spectroscopy analysis to investigate steroid production by differentiating hESCs. NMT conditioned

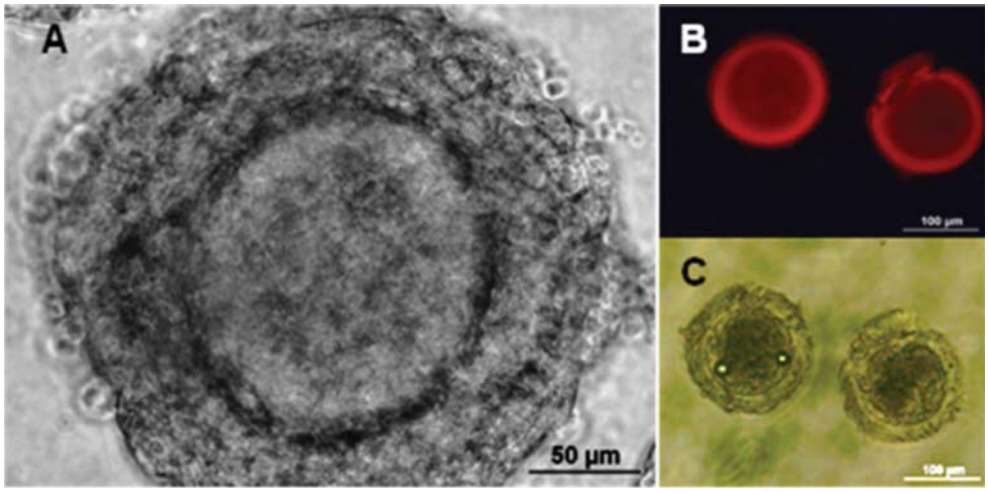


Figure 9 Phase contrast micrograph (**A**) of a structure within an embryoid body after 14 days of culture showing strong resemblance to an oocyte complex.

Monoclonal antibody localized zona pellucida on all stages of human oocytes (**B** and **C**) but always failed to detect zona-pellucida in these structures.

medium may also contain factors to induce steroid production which is required for gametogenesis. However, it was of interest that RA supplementation seems the most effective of these protocols in promoting post-meiotic germ cells, as determined by RNA qPCR analysis. To obtain further development of germ cells the optimal balance of these different factors will be important.

In conclusion, and in keeping with other studies, we have demonstrated that hESC lines can differentiate *in vitro* to a post-meiotic phenotype although this is a relatively ill-defined and inefficient process. This is perhaps not surprising as the germinal epithelium of either the seminiferous tubule of the developing testis or the follicle of the developing ovary is a highly complex stem cell niche *in vivo* and therefore difficult to replicate during EB differentiation in culture. Furthermore, whereas all the cell lines tested showed induction of some germ cells, the differences in the response of hESC lines to various growth factors means that there is considerable variation. Nevertheless, this *in vitro* differentiation allows investigation of early stages of human germ cell development not previously accessible. The aim in the near future is to develop more robust protocols, which are consistent with epigenetic, biochemical and functional development of germ cells for use in assays for drug and toxicology testing. Speculation on clinical applications for hESC-derived gametes is premature.

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